# iRpgFormatter for RDi

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## Overview

The iRpgFormatter formats the declarations, calculations, comments and embedded SQL statements of fully free-form (`**FREE`) RPGLE and SQLRPGLE sources.

Feature list:

- Indents the calculations by the nesting of their blocks and the comment lines like the following statement.
- Formats embedded SQL statements according to the RDi preferences page *Free-form SQL Formatting*.
- Formats the branches of conditional compiling (`/IF`, `/ELSE`, `/ENDIF`) like the compiler.
- Formats selected lines like the whole source.
- Various formatting options:
  - Casing style.
  - Add/remove const() keyword.
  - Break name on case change.
  - Sort const/value to end.
  - Replace dcl-pi name with *N.
  - Remove end-proc name.
  - Unindent compiler directives.
  - Outer parentheses of conditions (experimental).
  - Casing of operation codes and built-in functions (experimental).
  - Casing of special words of the calculations (experimental).
- Optionally formats the calculations with the IBM formatter.
- Optionally format source on save.

iRpgFormatter is an Open Source project that is hosted on github.com.

## Installation

iRpgFormatter requires RDi 9.6.0.13 or later. Install it with *Help* > *Install New Software...* from the update site or from the Eclipse Marketplace, see *iRpgFormatter On The Internet*.

## Usage

Right-click in the LPEX editor and select *Source* > *iFormat*. If lines are selected, only these lines are formatted, indented like in the whole source.

*Source* > *iFormat* is in the context menu of the *Remote Systems* view and of the *Project Explorer*, too:

- *Remote Systems*: object filters, libraries, source files and source members of type RPGLE or SQLRPGLE, IFS and local filters, folders and files.
- *Project Explorer*: folders and files with the extension RPGLE or SQLRPGLE.

There the sources are formatted in the background. At the end, a message lists the sources with errors or warnings.

To format with a key, map a user key action to the action `iRPGFormatter.Format` on the page *LPEX Editor* > *User Key Actions*.

With *Format on save*, a source is formatted, when you save it with *Save* or *Save All*.

## Preferences

The preferences page has been placed next to the IBM *Formatter* preferences page and is labeled as *iRpgFormatter*. The page *iRpgFormatter* at the root of the preferences leads there.

Play with the settings and watch the preview panel to understand the formatting options.

*Format declarations* formats the control options, the declarations and the compiler directives. *Format calculations* selects, who formats the calculations, the comment lines in front of them and the embedded SQL statements: *Off*, *IBM formatter* or *iRPGFormatter*.

With *IBM formatter*, the preview shows the calculations as iRPGFormatter would format them, because the IBM formatter cannot run in the preview.

Embedded SQL statements are formatted according to the RDi preferences page *Free-form SQL Formatting*, which is linked in the group *IBM Settings*. In RDi 9.6 they are only indented.

The following tabs contain the spelling of the words, that are formatted in the keyword casing style:

- Data types
- Declaration types
- Keywords
- Special Words
- Operation Codes
- Built-in Functions

The default entries of all lists are taken from the ILE RPG Reference for IBM i 7.6. The list Special Words is used for the declarations and the calculations.

Profiles exported with a former version can still be imported. Settings added later keep their current values.

A more detailed description of the settings can be found on the iRpgFormatter project page on GitHub.

Value "IBM formatter" of option "Format calculations" only works if the formatter is started in the Lpex editor.

**Limitations**

iRpgFormatter recognizes the statements and the blocks of a source, but it does not compile it. Therefore:

- `/COPY` and `/INCLUDE` members are not read. If a block starts or ends in such a member, the indentation may be wrong or the source cannot be formatted.
- If the branches of an `/IF` group end with different blocks, the formatter shows a warning, because the indentation of the following lines may be wrong.
- A statement, that cannot be formatted, e.g. because a line would exceed the end column, stays unchanged. A dialog lists these statements with their line numbers.

## Change Log 2.0.0.r (28.10.2026)

Plug-in: 9.6.0.13+

- Indents the calculations by the nesting of their blocks and the comment lines like the following statement.
- Formats embedded SQL statements according to the RDi preferences page *Free-form SQL Formatting*. In RDi 9.6 they are only indented.
- Formats the branches of conditional compiling like the compiler and warns, if they end with different blocks. Formats declarations with `/IF` between their lines section by section and leaves the lines after `/EOF` unchanged.
- Formats selected lines like the whole source.
- *Format declarations* and *Format calculations* replace *Execute iRPG formatter* and *Execute IBM formatter*. The former values are taken over.
- The calculations are now formatted by iRpgFormatter. To let the IBM formatter format them as before, set *Format calculations* to *IBM formatter*.
- New experimental settings: outer parentheses of conditions, casing of operation codes, built-in functions and special words of the calculations. New lists *Operation Codes* and *Built-in Functions*, more *Special Words*.
- Preference page *iRpgFormatter* at the root of the preferences. The IBM settings, that iRpgFormatter uses, are shown with links to their pages.

## Change Log 1.0.7.r (07.06.2026)

Plug-in: 9.6.0.13+

- Fix `Parents stack contains unexpected parent: dcl-ds...` on statements containing the `likerec` keyword.
- Fix incorrectly committed source code.

## Change Log 1.0.6.r (07.06.2026)

Plug-in: 9.6.0.13+

- Fix `Parents stack contains unexpected parent: dcl-ds...` on statements containing the `likerec` keyword.

## Change Log 1.0.5.r (01.06.2026)

Plug-in: 9.6.0.13+

- Fix `UnexpectedTokenException` when formatting keywords.

## Change Log 1.0.4.r (25.05.2026)

Plug-in: 9.6.0.13+

- Fix `ClassNotFoundException` in RDi 9.6.
- Applied experimental edge case fix: compiler conditions.
- Fix formatting compiler directives.
- Allow spaces between function name and parameter list.
- Fix setting maximum line length.

## Change Log 1.0.3.r (12.05.2026)

Plug-in: 9.6.0.13+

- Fix `Parents stack contains unexpected parent: dcl-pi *n end-pi;` parser error.
- Fix `Line n not found.` error in Lpex editor when positioning the cursor at the end of a formatting action.

## Change Log 1.0.2.r (12.05.2026)

Plug-in: 9.6.0.13+

- Fix `Unexpected token of type: FUNCTION` exception.

## Change Log 1.0.1.r (12.05.2026)

Plug-in: 9.6.0.13+

- Add formatting of compile time arrays.

## Change Log 1.0.0.r (12.05.2026)

Plug-in: 9.6.0.13+

- Initial version

# iRpgFormatter On The Internet

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| Project page on GitHub | https://github.com/tools-400/irpgformatter-plugin/ |
| Homepage | https://tools-400.github.io/irpgformatter/ |
| RDi 9.6+ Update Site | https://tools-400.github.io/irpgformatter/update-site/eclipse/rdi8.0/ |
| Eclipse Marketplace | https://marketplace.eclipse.org/content/irpgformatter-rdi-96013/ |
| RDi 9.6 Beta Update Site | https://tools-400.github.io/irpgformatter/beta-version/update-site/eclipse/rdi8.0/ |